

AI in Higher Education: Initial Teacher Training in the Critical and Didactic Use of Artificial Intelligence

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Abstract—This study was made possible by the University of La Laguna. The Educational Technology courses within the Bachelor's degrees in Early Childhood Education and in Physical Activity and Sports Sciences (n = 103) were redesigned, integrating AI agents such as ChatGPT, Gemini and Perplexity into six phases for the critical use of these tools—questioning, comparison, critical dialogue, verification, re-elaboration and reflection—which guided students' reflective interaction with AI. After the intervention, a mixed 23-item questionnaire was administered; descriptive statistics and thematic analysis of the "Evaluation and use of AI" dimension revealed that 88% of students believe AI should be didactically promoted in higher education and 42% identified ChatGPT as the agent providing the best answers. The most developed competencies were comparing outputs from different AI systems (66%), designing effective prompts (63%) and critically analysing responses (52%). The main potentialities highlighted were rapid access to information and time saving, while perceived risks centred on plagiarism and cognitive dependence. The findings corroborate the pedagogical validity of the model for strengthening prompt-engineering skills and critical thinking, yet underscore the need to deepen ethical training and rigorous verification frameworks to address concerns about academic integrity and AI reliability. It is concluded that a reflective and regulated integration of these technologies can significantly enhance teacher education in higher education.

Index Terms—Initial teacher training, artificial intelligence, ChatGPT, critical thinking, didactic model, higher education.

I. INTRODUCTION

IN RECENT years, artificial intelligence (AI) has burst into all areas of society, transforming everything from industrial processes to daily practices in education [1]. In this context, the university training of future professionals in education and sports cannot remain oblivious to this phenomenon, as rightly pointed out by [2], the university is responsible for linking learning with the requirements of the world of work. This article aims to show the development of an integration

and critical reflection model on the use of AI based on the work carried out by [3], which structures the didactic use of generative AI into six phases (interrogation, comparison, critical dialogue, verification, re-elaboration, and reflection) that guide a conscious, reasoned, and ethical interaction with these tools, as well as to know the students' opinions on the integration of AI in two subjects of the degrees of Teacher in Early Childhood Education and Physical Activity and Sports Sciences at the University of La Laguna.

To address this article, first, a review of the academic literature will be carried out to identify the most frequent uses of AI in university contexts, the existing pedagogical models that guide its incorporation from a critical perspective, and the students' opinions on its use. Later, the educational experience developed within the framework of an Innovation project at the University of La Laguna will be presented, in which a didactic model with a critical approach is applied to the previously mentioned degrees. Finally, as results, the evaluations and reflections of the participating students regarding this practice will be presented.

A. Use of AI in Higher Education

In recent years, artificial intelligence (AI) has gained increasing relevance in the field of higher education, generating a notable increase in scientific production aimed at exploring its pedagogical and technological applications. This interest has led to multiple studies that, from different disciplines, address the transformative potential of AI in university teaching and learning processes [4], [5].

One of the most representative investigations in this field is the systematic review carried out by [6], who analyzed the 50 most cited articles on AI in higher education. Their findings reveal that most studies have focused on predicting student academic performance and developing personalized learning support systems, with computer engineering being the predominant field of application. The authors emphasize the need to strengthen digital literacy for both faculty and students, while advocating for a critical approach from ethical and pedagogical perspectives.

Complementing this vision, [7] conducted a study on the scientific literature published between 2016 and 2022, with the purpose of systematizing the main uses of AI in university contexts. Their analysis broadens the research focus to include applications such as automated feedback, generation of

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teaching materials and questions, intelligent tutoring systems, virtual assistants, and learning management. Additionally, they identify a growing interdisciplinarity in the profile of researchers, mostly coming from faculties of education and computer science. In their conclusions, they highlight the accelerated expansion of AI as a key tool for personalizing, automating, and improving educational experience.

Another work to highlight for its connection to this study is the one carried out by [9], who delve into the use of generative AI (ChatGPT, Copilot, Gemini, Claude) in university environments with the main objective of analyzing how these tools can improve teaching and learning, promoting ethical and responsible use. Opportunities were identified (resource creation, more efficient planning, a boost to teacher productivity, and support for professional skills), challenges (ethics and responsible use, detection of AI-generated texts, and training in digital skills), as well as recommendations for their implementation (continuous training, fostering critical thinking, clear institutional policies, the development and use of specific prompts, as well as the monitoring and evaluation of learning). In the same vein, a review by [8] covering 59 articles published between 2019 and 2023 provides an updated view of research dynamics around AI in higher education. The study identifies China, Spain, and the United States as the countries with the highest scientific output, while noting limited international collaboration. Reported AI applications largely aim to foster deep learning by activating higher-order cognitive processes. Nevertheless, pedagogical guidance from instructors remains fundamental. The review concludes that, for an effective integration of AI in university teaching, institutions should invest in faculty training on these technologies and promote more collaborative and integrated research networks.

Finally, [10] reaffirm that existing research has paid special attention to general applications of AI, such as intelligent tutoring systems, adaptive environments, and learning analytics. However, they warn of important gaps that require priority attention in future research, such as the scarce incorporation of ethical perspectives, the limited presence of truly interdisciplinary approaches, and the need for greater methodological rigor. These shortcomings highlight the urgency of strengthening the theoretical and critical frameworks that support the integration of AI in higher education. Consequently, although current literature has advanced considerably in identifying the functional benefits of AI, it is essential to delve deeper into its pedagogical, social, and ethical implications. The consolidation of a more holistic perspective will not only allow leveraging the potential of these technologies to improve educational efficiency but also ensure a more equitable, critical, and common good-oriented university education.

B. Didactic Models for AI Integration

As proposed by [6] and [10], there is a gap in interdisciplinary approaches and a need for greater methodological rigor in the models or practices we carry out with AI. In recent years, different didactic models for integrating AI into initial training have been published with the intention of improving the previously described situation. Below (Table I),

TABLE I
DIDACTIC MODELS FOR AI INTEGRATION

Model	Proposal
TPACK Framework with AI [10]	Incorporates AI technological knowledge (e.g., AI-TK) and ethical aspects into the TPACK framework (technological-pedagogical-content knowledge).
AI Literacy Development Model for Teachers [11]	Didactic approach suggesting introducing future teachers to AI through guided experiences and case studies, combining activities with and without AI to develop their "AI literacy".
Case-Based Professional Development (Case-based PD) Model [12]	Consists of training teachers (in-service or pre-service) through practical cases where the integration of AI in teaching situations is simulated or analyzed.
Progressive Didactic Integration of AI Model [13]	Proposes a gradual adoption of AI in didactic processes: starting with simple automated tasks (e.g., automatic grading), then support tools (e.g., assistants for tutoring), and finally advanced systems integrated into the curriculum (e.g., class projects with AI).

after reviewing the main academic databases, those academic works grouped according to the identified didactic model are presented. It is important to note that these are models applied and evaluated in university classrooms involving faculty and students in the initial training of education professionals. Frameworks, guides, or strategies involving only faculty (except the TPACK model), as well as those not put into practice, have been discarded.

The didactic models presented in Table I are based on pedagogical approaches such as constructivism, problem-based learning (PBL), situated learning, and competency-based training. Technologically, they rely on intelligent tutoring systems, learning analytics, adaptive platforms, and generative tools like ChatGPT. It is essential that faculty not only use them competently but also understand the mechanisms, potential, and limits of these technologies, thus developing a critical and reflective view on their educational application. Initially, models with practical application in university classrooms (faculty and students) are presented, followed by the Intelligent-TPACK model, in which only teachers participated. Reference [12] does not propose a structured and coded model but does articulate a structured pedagogical approach or proposal. The article shows several examples carried out in university classrooms with the following steps: purpose of using AI, objectives, example, instruction for AI, follow-up activities, potential results, potential challenges, questions for reflection. Similarly, [13] does not propose a formal model with its own name. However, it presents a structured approach for teacher professional development in the context of artificial intelligence (AI), based on the use of case studies and collaborative discussions. The components of this approach are: Direct instruction on AI, structured and unstructured case studies, collaborative discussion (faculty), and practical application. Lastly, [14] proposes the following: awareness, specific training, practical application, teacher collaboration, and continuous evaluation and improvement.

The results of applying these approaches by [12], [13], and [14] coincide in pointing out that artificial intelligence (AI) represents a tool with great potential to transform teacher training, especially in terms of learning personalization, atten-

tion to diversity, and strengthening critical thinking. In general, pre-service teachers perceive AI positively and recognize its pedagogical value, highlighting that it can improve planning, reflection, and creativity in the classroom. However, common barriers such as lack of specific training, insufficient technological infrastructure, and some resistance to change are also identified. Despite these challenges, the studies propose effective strategies for meaningful implementation. Reference [13] show that the combination of direct instruction and case-based discussion improves AI literacy and promotes critical reflection. For their part, [12] highlight the utility of integrating practical and ethical activities that prepare future teachers for a conscious pedagogical use of these technologies. Finally, [14] underscores the need to design continuous training programs, foster communities of practice, and have institutional support as keys to overcoming barriers and moving towards an ethical and transformative integration of AI in education.

The only model that proposes a formal structured model is the extension of the TPACK model to AI [11]. This suggests that faculty should not only develop technical skills but also pedagogical understanding and ethical judgment for its implementation in the classroom. To this end, they develop a model called Intelligent-TPACK which considers the following issues: knowledge of artificial intelligence (AI-Knowledge), ethical, technological, pedagogical knowledge (Ethical-TPEK), and critical and reflective vision. It should be noted that this model was only evaluated on university teachers, obtaining the following results:

- Technological and pedagogical knowledge strongly influences the ability to integrate AI effectively.
- Teachers with more ethical and reflective training showed greater awareness of the limits and risks of using AI.
- The Intelligent-TPACK model proved valid and coherent for evaluating the teaching competencies needed in the age of AI.

The four reviewed approaches coincide in recognizing the enormous transformative potential of generative AI for teacher training, but they reveal tensions between the urgency to innovate and the real limitations of the context. While [12], [13], and [14] offer flexible practical guides—focused on classroom examples, case studies, and cycles of continuous improvement—they lack a formal codification that facilitates their replication and rigorous evaluation. In contrast, the Intelligent-TPACK model [11] provides a validated structure that integrates pedagogical, technological, AI, and ethical knowledge, although it has only been applied to university faculty. This duality between flexibility and robustness is exacerbated by persistent gaps: a lack of specific training, unequal infrastructure, limited student input, and ethics being treated more as a matter of reflection than as an operational protocol.

The method section will present and describe the didactic model for the critical use of AI proposed by [3] and applied in this experience. The final section of the introduction will now be presented, compiling studies that have gathered student opinions regarding the use of AI.

C. Student Opinions on the Use of AI in University Classrooms

In this section, we would like to gather the evaluations made by students regarding the use of AI in initial teacher training. Several recent investigations have explored the perceptions of pre-service teacher students regarding the integration of AI in educational contexts. Reference [15] highlight the growing interest of students in acquiring critical digital competencies, identifying both the pedagogical potential of AI and the need for specific training for its ethical and didactic use. Along the same lines, [16], based on a study conducted in Hong Kong, show a positive assessment of generative AI, especially for its ability to offer immediate feedback, generate ideas, and personalize learning, although they also warn of ethical and academic concerns. For their part, [17] analyze how students in the educational field begin to incorporate AI tools into their study and teaching processes, emphasizing the importance of accompanying their implementation with critical digital literacy. Reference [18] investigated the use of generative AI in lesson planning by pre-service teachers, noting low practical integration despite recognizing its usefulness. This highlights the need for more structured training strategies. Finally, [19] analyzed how AI contributes to the development of critical thinking, self-assessment, and improved autonomous learning in university students. Collectively, these studies agree that students' attitudes toward AI are positive, but they emphasize the urgent need to strengthen didactic, ethical, and training frameworks for its effective and responsible incorporation into initial teacher education.

As a final point, [20] analyzed the impact of using AI tools on critical thinking and collaborative work in higher education. Using a quantitative methodology with surveys applied to 465 university students in Ecuador, the authors identified a predominantly positive attitude towards these technologies. The results indicate that 64% of students reported improvements in their critical thinking, while 59% reported that these tools facilitated teamwork. The greatest benefits were related to: idea generation, automated feedback, and improved argumentation skills, while the main concerns focused on ethical issues such as plagiarism, the veracity of responses, and the risk of technological dependence.

As we have seen, the perceptions of students in initial teacher training regarding the reflective and critical use of artificial intelligence reflect a growing consensus: AI possesses high pedagogical potential to strengthen digital competencies, creativity, and metacognition, provided its integration is guided by rigorous ethical frameworks and deep technological literacy. This aligns with recent research that highlights the need for AI digital literacy and prompt engineering (i.e., how we formulate precise and effective inputs or questions to AI) as foundations for critical thinking [21]. Other studies indicate that students value the instant and personalized feedback offered by generative tools [22], although they express reservations about academic integrity and cognitive dependence. Furthermore, the findings of [20] in Ecuador, with 64% of students recognizing improvements in critical thinking and 59% in collaboration thanks to the use of AI, directly align

with reviews that advocate for reconfiguring teacher training to incorporate critical evaluation, self-regulation, and self-directed learning environments [23].

The experience developed in the degrees of Early Childhood Education and Physical Activity-Sports Sciences at the University of La Laguna will now be presented.

II. METHOD

This study seeks to investigate the perceptions of students in Education degrees (Degree in Early Childhood Education Teacher and Degree in Physical Activity and Sports Sciences (CAFyD)) regarding the incorporation of Artificial Intelligence (AI) in their initial teacher training. As a fundamental premise, AI-based tools were integrated into the Educational Technology subjects and proposed tasks. The pedagogical approach followed the didactic model for the development of critical thinking with AI in higher education, proposed by [3]. In this vein, the following research question arises: What are the perceptions of students (future educators) subsequent to the implementation of a critical model for AI integration?

The research design first involved an exhaustive review of the critical thinking model with AI. Based on this framework, the learning activities in the Educational Technology subjects of the target degrees were redesigned, reorganizing the academic planning to promote rational, visible, and reflective interaction with different Artificial Intelligence tools. After its implementation in the classroom and once the subjects were completed, a questionnaire with closed and open-ended questions was used to collect students' perceptions about the relevance and usefulness of AI as learning support, as well as their stance on the challenges posed by the integration of these technologies.

A. Integration of the Didactic Model for the Development of Critical Thinking With Artificial Intelligence in Higher Education

The proposal by [3] focuses on reconceptualizing learning tasks and their execution process, supported by a critical use of AI tools such as ChatGPT, Gemini, or Copilot. The aim is for students to use AI reflectively and transparently, motivating the elaboration of reflection tasks and learning products. To this end, the author establishes six training phases or dimensions (Fig. 1):

- **Interrogation:** In the first phase, student motivation is awakened, and the elaboration of clear and relevant prompts (questions or instructions) is encouraged. The objective is to teach "prompt engineering" so that the use of AI is precise and productive, generating quality responses.
- **Comparative Application:** In this second phase, students use the same instruction (prompt) in various AI tools to compare the responses obtained and analyze how each agent (e.g., ChatGPT, Gemini, Perplexity, or others) processes information differently. This exercise promotes awareness of potential biases and the specific limitations of each system.

- **Critical Dialogue:** Once the responses are compared, students maintain active interaction with the AI, reformulating questions, asking for clarifications, and contrasting arguments. Thus, a continuous dialogue is promoted that helps deepen understanding of the topic.
- **Data Verification:** During the fourth phase, the importance of externally corroborating the information generated by AI is emphasized. To do this, students resort to additional sources, bibliography, or consultations with faculty, so they can evaluate the accuracy and relevance of the obtained content.
- **Personal Re-elaboration:** This fifth phase constitutes an essential pillar of the model, as it requires the student to rewrite and assimilate the information in their own words. The purpose is to integrate the ideas provided by AI into an original narrative that reflects the personal perspective, avoiding literal reproduction and enhancing reflective writing and argumentation.
- **Reflection:** The final phase consists of the student conducting a critical analysis of the entire process and drawing conclusions about their experience with AI. They are expected to describe the learning achieved, the obstacles encountered, and future possibilities for integrating these technologies formatively in their academic and professional field.

For the concrete implementation of these phases, the model proposes that the faculty first present the academic task and clearly define the expected final product and the evaluation criteria. Students then design prompts to request information from different AI agents and compare the responses, deepening the dialogue with the tool. After verifying the information, they must produce an original version of the content and accompany it with documentation of the process and their personal reflection, evidencing their evolution through the six described phases. Evaluation considers both the final product and the quality of interaction with AI, the relevance of consulted sources, and the degree of critical awareness demonstrated by the student. The model emphasizes a paradigm shift in university teaching, where AI becomes an ally to promote competencies of inquiry, verification, dialogue, and re-elaboration, guiding the formative process towards an integral development of critical thinking [3].

B. Sample

This study was conducted with first-year students from the degrees of Early Childhood Education Teacher and Physical Activity and Sports Sciences (CAFyD) at the University of La Laguna during the 2024-2025 academic year. Two groups from the Educational Technology subject (one for each degree) were selected, forming a total of 103 participants ($N = 103$). All participated voluntarily after being informed about the research objectives and the guarantee of confidentiality, using their data exclusively for academic and teaching improvement purposes. The sample selection followed a convenience criterion, a non-probabilistic modality based on student accessibility and willingness. Ease of access to these groups, as well as the temporal coincidence with the end of the first semester of the

2024–2025 academic year (end of December 2024), favored the configuration of this sample. Consequently, the internal validity and representativeness of the results are limited to the context in which the study's researchers teach. Therefore, the sample is conditioned by the specific circumstances of the participating groups. Furthermore, as this is a pre-post design without a control group, the presence of uncontrolled extraneous variables that may have contributed to the observed changes cannot be ruled out.

Regarding external validity, while the findings offer relevant insights into the effectiveness of a critical model for AI integration in initial teacher training, their applicability to other educational contexts or different training programs must be interpreted with caution. Factors such as prior training in digital competencies, familiarity with AI tools, or curricular differences between degree programs could significantly influence the results if the study were to be replicated in other settings.

C. Instrumentation

The instrument was created *ad hoc* within the framework of an innovation and educational transfer project supported by the Vice-Rectorate for Teaching Innovation and Quality of the University of La Laguna. For its validation, an expert panel consisting of faculty from the areas of Research Methods and Diagnosis in Education, and Didactics and School Organization was consulted, following recommendations by [24] to ensure content validity.

The final questionnaire included 23 items of different types (Likert scale, multiple-choice, and open-ended short-answer questions) following the guidelines of [25] for designing mixed instruments. These items were organized into 8 dimensions of analysis: motivation with the subject, frequency and hours of connection; conceptual learning; specific digital competencies acquired; attitudes and uses of artificial intelligence; satisfaction with the methodology and virtual classroom; level of content demand; adequacy of practices, and qualitative feedback and improvement proposals. This work focuses on the specific dimension of assessment and use of Artificial Intelligence, composed of four items designed to capture attitudes, experiences, and learning after using AI tools in the subject. The first item (multiple choice) inquires about students' opinions on the utility of AI in education (e.g., whether it favors university teaching or encourages plagiarism), allowing selection of multiple positions simultaneously. The second item (single choice) requests a comparative satisfaction rating with three specific AI systems (ChatGPT, Gemini, and Perplexity, in this case), choosing which provided the best responses. Next, the following item (multiple choice) explores learning derived from using AI, such as designing more effective prompts, comparing results between tools, and critically re-elaborating received information. Finally, the last item (short-answer question) openly collects perceived potentialities and negative aspects in the application of AI to the teaching-learning process. This combination of quantitative and qualitative formats aims to capture a view of how students value, use, and reflect on AI technologies in their training context.

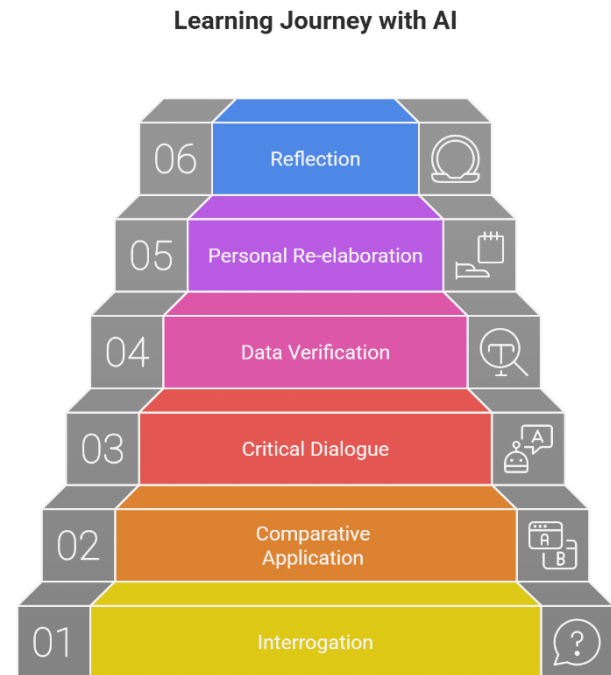


Fig. 1. Phases of the model for developing critical thinking with AI. Own elaboration.

D. Data Analysis

The study followed a mixed-methods approach integrating a non-experimental descriptive design (for quantitative items) and a qualitative analysis of open-ended responses. In the quantitative aspect, absolute and percentage frequencies were calculated using SPSS, as proposed by [26] to ensure consistency in presenting results. This procedure allowed for precise description of assessments about AI, comparative satisfaction in implementing different agents, and learning derived from using AI tools in the teaching and learning process. In parallel, the qualitative analysis of open-ended responses to the item about opportunities and risks of AI was performed using thematic analysis following the recommendations of [27] for qualitative coding. Coding was structured in three stages: generation of initial codes, systematic application of these codes to all responses; and review and refinement, incorporating emerging low-frequency codes. To ensure reliability, double coding was performed on 20% of the responses, and consensus was reached through discussion in cases of discrepancy.

III. RESULTS

The descriptive results for the four items of the Assessment and use of AI dimension are presented below, both jointly and disaggregated by group.

A. Opinions on the Utility of AI

In the combined analysis, for the item After taking the course, with which opinions about the utility of AI (Artificial Intelligence) in education do you agree? You can mark several answers, the most selected option was “AI is useful for students (preparing assignments, creating resources, searching

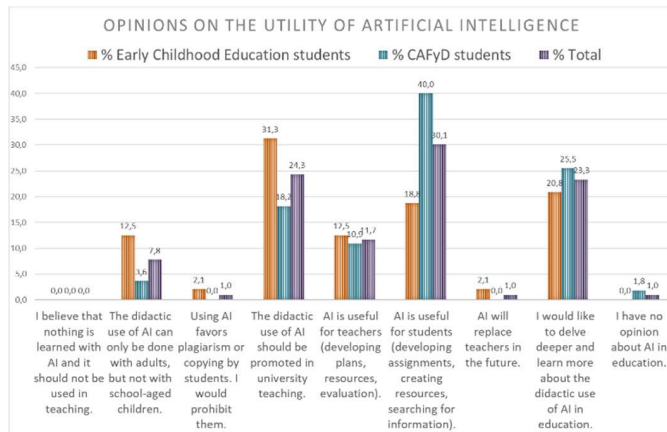


Fig. 2. Opinions on the utility of artificial intelligence in teaching and learning.

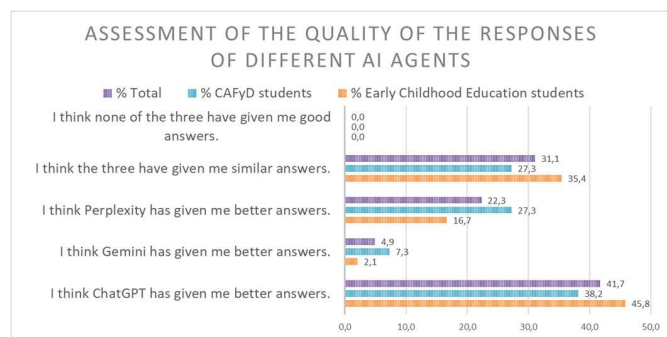


Fig. 3. Assessment of the quality of responses from different artificial intelligence agents.

for information)” (31 responses; 30.1%), followed by “The didactic use of AI should be promoted in university teaching” (25; 24.3%) and “I would like to deepen and learn more about the didactic use of AI in education” (24; 23.3%). 11.7% highlighted its usefulness for teachers, and only 7.8% believed it could only be used with adults. No participant considered that AI contributed nothing or expressed a neutral opinion (Fig. 2).

By group, Early Childhood Education students were more in favor of promoting its use in university teaching (31.3%). In contrast, in CAFyD, 40.0% particularly valued the utility of AI for students.

B. Comparative Satisfaction Regarding the Quality of Responses From AI Agents

During the application of the model and the course of the subject, in several practical tasks, students used three artificial intelligence agents comparatively: ChatGPT, Gemini, and Perplexity. In the item Rate the degree of satisfaction with the experience of using the three AIs in this subject, ChatGPT was chosen as the tool that offered the best responses by 41.7% of participants (43/103), followed by Perplexity (22.3%; 23/103) and Gemini (4.9%; 5/103). 31.1% considered the three platforms equivalent, and no respondent showed total dissatisfaction (Fig. 3).

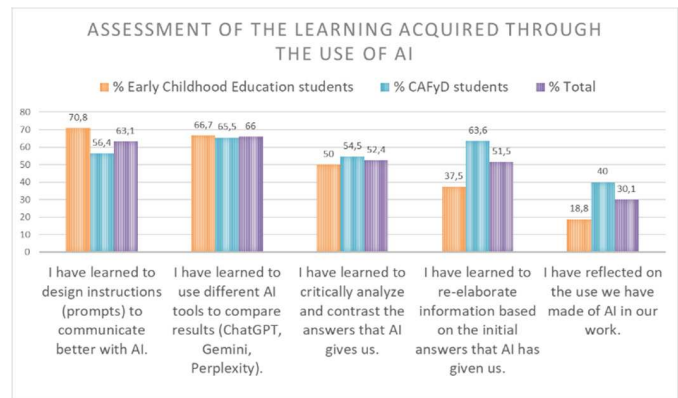


Fig. 4. Assessment of learning acquired when using artificial intelligence in the practical tasks proposed in the subjects.

In the case of Early Childhood Education Teacher students, 45.8% preferred ChatGPT, and 35.4% valued the similarity between the three; in CAFyD, ChatGPT reached 38.2%, and Perplexity 27.3%.

C. Learning Derived From the Use of AI

In the item What have you learned after carrying out the activities in which you had to use AI?, the combined analysis revealed that the most developed competencies were, in this order, comparing results between different AI tools (68 students marked this option; 66.0%), designing effective instructions or “prompts” (65; 63.1%), and critically analyzing the responses generated by AI (54; 52.4%) (Fig. 4). Likewise, 51.5% of the sample indicated having learned to re-elaborate information based on initial responses, and 30.1% reported having reflected on their own use of these technologies.

Significant differences stand out between the groups: Early Childhood Education students reported more frequently having improved in designing prompts (70.8% vs. 56.4% in CAFyD), while the CAFyD group reported critical re-elaboration of information to a greater extent (63.6% vs. 37.5% in Early Childhood Education). These results underscore that, although both groups recognize the pedagogical value of AI, there is some variability in the specific skills acquired depending on the degree profile.

D. Reflections on Potentialities and Risks of Using AI in University Training

The thematic analysis of the 101 valid responses revealed eight benefits and five risks that shape students’ perception of the use of AI in university teaching (Table II).

Among the potentialities, the most reiterated was agile and expanded access to information (45 mentions; 44.6%), especially noted by CAFyD students (57.4%), highlighting that it “facilitates finding updated materials for planning sessions”. This was followed by time savings and task automation (16.8%), and the generation of creative ideas (16.8%), valued as an “enhancer for creating innovative resources”. The Early Childhood Education Teacher degree group also emphasized learning personalization (8.5%), stressing the possibility of

TABLE II
ASSESSMENT OF THE POTENTIALITIES AND RISKS OF
USING AI IN UNIVERSITY TEACHING

Thematic code	ECE Students (n = 47)	CAF Students (n = 54)	Total	% E CE	% CAF	% Total
Access and search information	14	31	45	29,8	57,4	44,6
Risk of plagiarism/copying	9	16	25	19,1	29,6	24,8
Efficiency/time saving	8	9	17	17,0	16,7	16,8
Creativity and inspiration	6	11	17	12,8	20,4	16,8
Dependence/overconfidence	7	7	14	14,9	13,0	13,9
Reliability	5	8	13	10,6	14,8	12,9
Dehumanization/loss interaction	6	6	12	12,8	11,1	11,9
Personalization/adaptive learn.	4	6	10	8,5	11,1	9,9
Privacy and data protection	3	5	8	6,4	9,3	7,9

adapting content “to each student’s pace”. Moreover, efficiency and time-saving in completing tasks were positively valued (“It can help you do various tasks faster”), as was its capacity to stimulate creativity in the initial stages of work: “It can give you ideas you’d never think of.”

Regarding perceived risks, the primary concern was plagiarism or copying academic work (24.8%), with a greater emphasis in CAFyD (29.6%) than in Early Childhood Education (19.1%). Students warned that “if not supervised, students might be tempted to submit assignments entirely generated by AI.” The second fear alluded to excessive dependence on the tool (13.9%), linked to “the loss of personal effort and critical thinking.” Other recurring concerns included the dehumanization of teaching (“We would lose human skills necessary in our daily lives.”) at 11.9% (due to the potential replacement of teacher-student interaction) and the reliability of content (12.9%), reflected in comments about the “bias” or “errors” AI can offer: “AI allows us to have a great deal of information at our fingertips [...]. However, it can also make mistakes, so in many cases, we shouldn’t fully trust its answers.” Although less frequently cited, data privacy (7.9%) emerged as a technical-ethical risk. These results allow us to understand how students perceive an ambivalent balance: they recognize AI as a tool that accelerates information search and fosters creativity, but express ethical-pedagogical reservations regarding plagiarism, cognitive dependence, and reliability. These perceptions present nuances by degree: CAFyD students emphasize the benefits of access and creativity, while Early Childhood Education highlights personalization and expresses more caution regarding the dehumanization of the educational process.

IV. DISCUSSION AND CONCLUSION

Didactic models for AI integration are quite nascent, requiring ongoing application and evaluation to refine aspects identified through classroom observation and feedback from educators implementing them. Our experience with the model proposed by [3] has been positive, corroborated by student surveys where a majority affirm the need to promote the

didactic use of AI in university teaching. This aligns with other studies, such as [15], which highlight students’ growing interest in acquiring critical digital competencies, recognizing both AI’s pedagogical potential and the need for specific training in its ethical and didactic use.

Another important point is the structured steps offered by the various models discussed here. All proposed approaches provide a didactic sequence for working with AI. Two aspects stand out most in all analyzed models ([9], [11], [12], [13], [14], and [3]): first, the purpose or interrogation of using AI, and second, the necessity of incorporating reflection into any process undertaken with these tools. This suggests that most authors agree on the importance of establishing a didactic framework that includes these crucial elements.

Regarding student evaluations of AI use, its importance for accessing and searching for information is most notable, making it the most frequently cited item. This suggests that conversational bots have become a powerful tool for students. This finding is consistent with [14], [16], and [20], who identify how students in the educational sphere are beginning to incorporate AI tools into their study and teaching processes.

Among the perceived risks, a significant concern is students’ worry about the loss of personal effort and critical thinking. This aligns with studies by [6], [10], [11], [15], [16], [17], and [20], who also emphasize the need to reinforce this aspect through critical AI literacy. This contrasts with the work of [12], [13], and [14], who believe that AI use has strengthened critical thinking.

Another highly perceived risk is the potential for students to plagiarize in their assignments. It’s significant that students themselves express this concern, underscoring the greater need for a didactic model of AI integration that incorporates these technologies into the teaching-learning processes we conduct with our students. This also aligns with the majority of studies, which stress the importance of using these tools ethically ([15], [17], [16], [11], [10], [17], and [20]).

This work is merely the beginning of a series of investigations that the scientific community must undertake in the coming years. This involves incorporating and applying didactic models with a critical perspective of AI in university classrooms and sharing the results with colleagues to improve teaching-learning processes in higher education.

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REFERENCES

- [1] O. C. García, “Inteligencia artificial en educación superior: Oportunidades y riesgos,” *RiiTE Revista Interuniversitaria de Investigación en Tecnología Educativa*, vol. 15, no. 15, pp. 16–27, 2023, doi: [10.6018/riite.591581](https://doi.org/10.6018/riite.591581).

- [2] S. S. M. Gutierrez, S. L. Pérez, and M. G. Munguía, "Artificial intelligence in e-learning plausible scenarios in Latin America and new graduation competencies," *IEEE Revista Iberoamericana de Tecnologías del Aprendizaje*, vol. 17, no. 1, pp. 31–40, Feb. 2022, doi: [10.1109/RITA.2022.3149833](https://doi.org/10.1109/RITA.2022.3149833).
- [3] M. Area-Moreira, "Luces y sombras de la IA en la educación superior. Didáctica para el pensamiento crítico," RIULL Inst. Repository Univ. La Laguna, Santa Cruz de Tenerife, Spain, Tech. Rep., 2025. [Online]. Available: <http://riull ull.es/xmlui/handle/915/40470>
- [4] A. M. Al-Zahrani and T. M. Alasmari, "Exploring the impact of artificial intelligence on higher education: The dynamics of ethical, social, and educational implications," *Humanities Social Sci. Commun.*, vol. 11, no. 1, Jul. 2024, Art. no. 912, doi: [10.1057/s41599-024-03432-4](https://doi.org/10.1057/s41599-024-03432-4).
- [5] G. P. Chávez Granizo, J. K. Castro Game, M. A. I. Martínez, and Y. F. Tobar Flores, "La inteligencia artificial en la educación superior: Oportunidades y amenazas," *Reciamuc*, vol. 8, no. 1, pp. 71–79, Jan. 2024, doi: [10.26820/reciamuc/8.\(1\).ene.2024.71-79](https://doi.org/10.26820/reciamuc/8.(1).ene.2024.71-79).
- [6] H. C. Chu, G. H. Hwang, Y. F. Tu, and K. H. Yang, "Roles and research trends of artificial intelligence in higher education: A systematic review of the top 50 most-cited articles," *Australas. J. Educ. Technol.*, vol. 38, no. 3, pp. 22–42, 2022, doi: [10.14742/ajet.7526](https://doi.org/10.14742/ajet.7526).
- [7] H. Crompton and D. Burke, "Artificial intelligence in higher education: The state of the field," *Int. J. Educ. Technol. Higher Educ.*, vol. 20, no. 1, p. 22, Apr. 2023, doi: [10.1186/s41239-023-00392-8](https://doi.org/10.1186/s41239-023-00392-8).
- [8] O. L. Regalado, N. Núñez-Rojas, O. R. L. Gil, and J. Sánchez-Rodríguez, "El Análisis del uso de la inteligencia artificial en la educación universitaria: Una revisión sistemática (Analysis of the use of artificial intelligence in university education: A systematic review)," *Pixel-Bit Rev. Medios Educ.*, vol. 70, no. 70, pp. 97–122, 2024, doi: [10.12795/pixelbit.106336](https://doi.org/10.12795/pixelbit.106336).
- [9] J. Cordero and A. Cordero-Castillo, "Exploring the potential of generative AI in education: Opportunities, challenges, and best practices for classroom integration," in *Foundations of Computer Science and Frontiers in Education: Computer Science and Computer Engineering (CSCE)* (Communications in Computer and Information Science), vol. 2261, H. R. Arabnia, L. Deligiannidis, S. Amirian, F. G. Mohammadi, and F. Shenavarmasouleh, Eds., Cham, Switzerland: Springer, 2024, pp. 252–266, doi: [10.1007/978-3-031-85930-4_23](https://doi.org/10.1007/978-3-031-85930-4_23).
- [10] M. Bond et al., "A meta systematic review of artificial intelligence in higher education: A call for increased ethics, collaboration, and rigour," *Int. J. Educ. Technol. Higher Educ.*, vol. 21, no. 1, Jan. 2024, Art. no. 4, doi: [10.1186/s41239-023-00436-z](https://doi.org/10.1186/s41239-023-00436-z).
- [11] I. Celik, "Towards intelligent-TPACK: An empirical study on teachers' professional knowledge to ethically integrate artificial intelligence (AI)-based tools into education," *Comput. Hum. Behav.*, vol. 138, Jan. 2023, Art. no. 107468, doi: [10.1016/j.chb.2022.107468](https://doi.org/10.1016/j.chb.2022.107468).
- [12] V. Alexandrowicz, "Artificial intelligence integration in teacher education: Navigating benefits, challenges, and transformative pedagogy," *J. Educ. Learn.*, vol. 13, no. 6, p. 346, Nov. 2024, doi: [10.5539/jel.v13n6p346](https://doi.org/10.5539/jel.v13n6p346).
- [13] A.-C.-E. Ding, L. Shi, H. Yang, and I. Choi, "Enhancing teacher AI literacy and integration through different types of cases in teacher professional development," *Comput. Educ. Open*, vol. 6, Jun. 2024, Art. no. 100178, doi: [10.1016/j.caeo.2024.100178](https://doi.org/10.1016/j.caeo.2024.100178).
- [14] M. E. Mera-Alvarado, "Integration of artificial intelligence in teacher training," *Polo Conocimiento*, vol. 10, no. 2, pp. 347–359, 2025, doi: [10.23857/pc.v10i2.8878](https://doi.org/10.23857/pc.v10i2.8878). [Online]. Available: <https://polodelconocimiento.com/ojs/index.php/es/article/view/8878/html>
- [15] D. Ayuso-del Puerto and P. Gutiérrez Esteban, "La inteligencia artificial como recurso educativo durante La formación inicial del profesorado," *RIED-Revista Iberoamer. de Educación a Distancia*, vol. 25, no. 2, pp. 347–362, 2022, doi: [10.5944/ried.25.2.32332](https://doi.org/10.5944/ried.25.2.32332).
- [16] C. K. Y. Chan and W. Hu, "Students' voices on generative AI: Perceptions, benefits, and challenges in higher education," *Int. J. Educ. Technol. Higher Educ.*, vol. 20, no. 1, p. 43, Jul. 2023, doi: [10.1186/s41239-023-00411-8](https://doi.org/10.1186/s41239-023-00411-8).
- [17] J. F. Álvarez-Herrero, "Opinión del alumnado universitario de educación sobre el uso de la IA en sus tareas académicas," *Eur. Public Social Innov. Rev.*, vol. 9, pp. 1–18, 2024, doi: [10.31637/epsir-2024-534](https://doi.org/10.31637/epsir-2024-534).
- [18] H. P. Hsu et al., "Preliminary Study on pre-service teachers' applications and perceptions of generative artificial intelligence for lesson planning," *J. Technol. Teacher Educ.*, vol. 32, no. 3, pp. 409–437, 2024, doi: [10.70725/897776dkibzu](https://doi.org/10.70725/897776dkibzu).
- [19] Z. Slimi, A. Benayoune, and A. E. Alemu, "Students' perceptions of artificial intelligence integration in higher education," *Eur. J. Educ. Res.*, vol. 14, no. 2, pp. 471–484, Mar. 2025, doi: [10.12973/eu-jer.14.2.471](https://doi.org/10.12973/eu-jer.14.2.471).
- [20] L. I. Ruiz-Rojas, L. Salvador-Ullauri, and P. Acosta-Vargas, "Collaborative working and critical thinking: Adoption of generative artificial intelligence tools in higher education," *Sustainability*, vol. 16, no. 13, p. 5367, 2024, doi: [10.3390/su16135367](https://doi.org/10.3390/su16135367).
- [21] Y. Walter, "Embracing the future of artificial intelligence in the classroom: The relevance of AI literacy, prompt engineering, and critical thinking in modern education," *Int. J. Educ. Technol. Higher Educ.*, vol. 21, p. 15, Feb. 2024, doi: [10.1186/s41239-024-00448-3](https://doi.org/10.1186/s41239-024-00448-3).
- [22] S. Haroud and N. Saqri, "Generative ai in higher education: Teachers' and students' perspectives on support, replacement, and digital literacy," *Educ. Sci.*, vol. 15, no. 4, p. 396, 2025, doi: [10.3390/educsci15040396](https://doi.org/10.3390/educsci15040396).
- [23] Q. Xia, X. Weng, F. Ouyang, T. Lin, and T. Chiu, "A scoping review on how generative artificial intelligence transforms assessment in higher education," *Int. J. Educ. Technol. Higher Educ.*, vol. 21, p. 40, May 2024, doi: [10.1186/s41239-024-00468-z](https://doi.org/10.1186/s41239-024-00468-z).
- [24] C. Teddlie, R. B. Johnson, and A. Tashakkori, *Foundations of Mixed Methods Research*, 2nd ed. Thousand Oaks, CA, USA: SAGE, 2021.
- [25] J. W. Creswell and J. D. Creswell, *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches*, 5th ed. Thousand Oaks, CA, USA: SAGE, 2018.
- [26] G. P. G. Alban, A. E. V. Arguello, and N. E. C. Molina, "Metodologías de investigación educativa (descriptivas, experimentales, participativas, y de investigación-acción)," *Recimundo*, vol. 4, no. 3, pp. 163–173, Jul. 2020, doi: [10.26820/recimundo/4.\(3\).julio.2020.163-173](https://doi.org/10.26820/recimundo/4.(3).julio.2020.163-173).
- [27] J. Saldaña, *The Coding Manual for Qualitative Researchers*, 4th ed. Thousand Oaks, CA, USA: SAGE, 2021.